(10,10): three exact blocks and a separated sphere redraw

11 recorded steps | 13 individual operations | 24 external adjacent checks

Use the original EXACT algebraic A, B and C source blocks imported from the other project. Each has type(4,3) and product D5^2. Assemble B,A,C explicitly; this is a three-copy construction. Independent seven-factor rotations share one SAME input. Every subsequent source pull and both matching-half combines are checked before the sixth point is added on the LEFT.

The complete embedded target remains recorded. D5^6 is never assumed central in B6. The actual merged pair matches the saved local chart individually. First redraw ONLY the same pair curve from its complementary side, retaining factor identity, power, full disk discrepancy and separate sphere/+I checks. The following cabled half-lantern substitution is an EXACT DISK relation with its own complete named proof.

The final20 actual factors have type(10,10), identity marked-sphere action and chosen+I lift. Four later drawing edits in the old project fail closure; their exact data and nonidentity homology matrix are preserved separately. They are not used here. The older twenty dictated routes in this repository remain unchanged.

Reading the diagrams. Half-twist arcs have H labels; closed curves have T labels. Ordinary triple twists are temporary calculation factors. All factor grids have SIX UNNUMBERED points at18pt pitch and1.26pt radius. The connected braid represents the exact displayed factor words. Individual supports are not a joint cut-system or jointly minimal drawing.

Certificate scope. Faithful disk actions, raw free-word reconstructions, complete actual kernels AND shadows when the last strand is fixed, full actions for halves that move it, and separate sphere/lift checks remain in 10-10.json. Internal named-rule checks are retained separately from the external-check count. No unchosen framed-half-hole convention is asserted.

The explicit starting construction

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

1. four_three_B.1
T3^2

2. four_three_B.2
T3^2

3. four_three_B.3
H

4. four_three_B.4
T3^2

5. four_three_B.5
H

6. four_three_B.6
H

7. four_three_B.7
H

8. four_three_A.1
T3^2

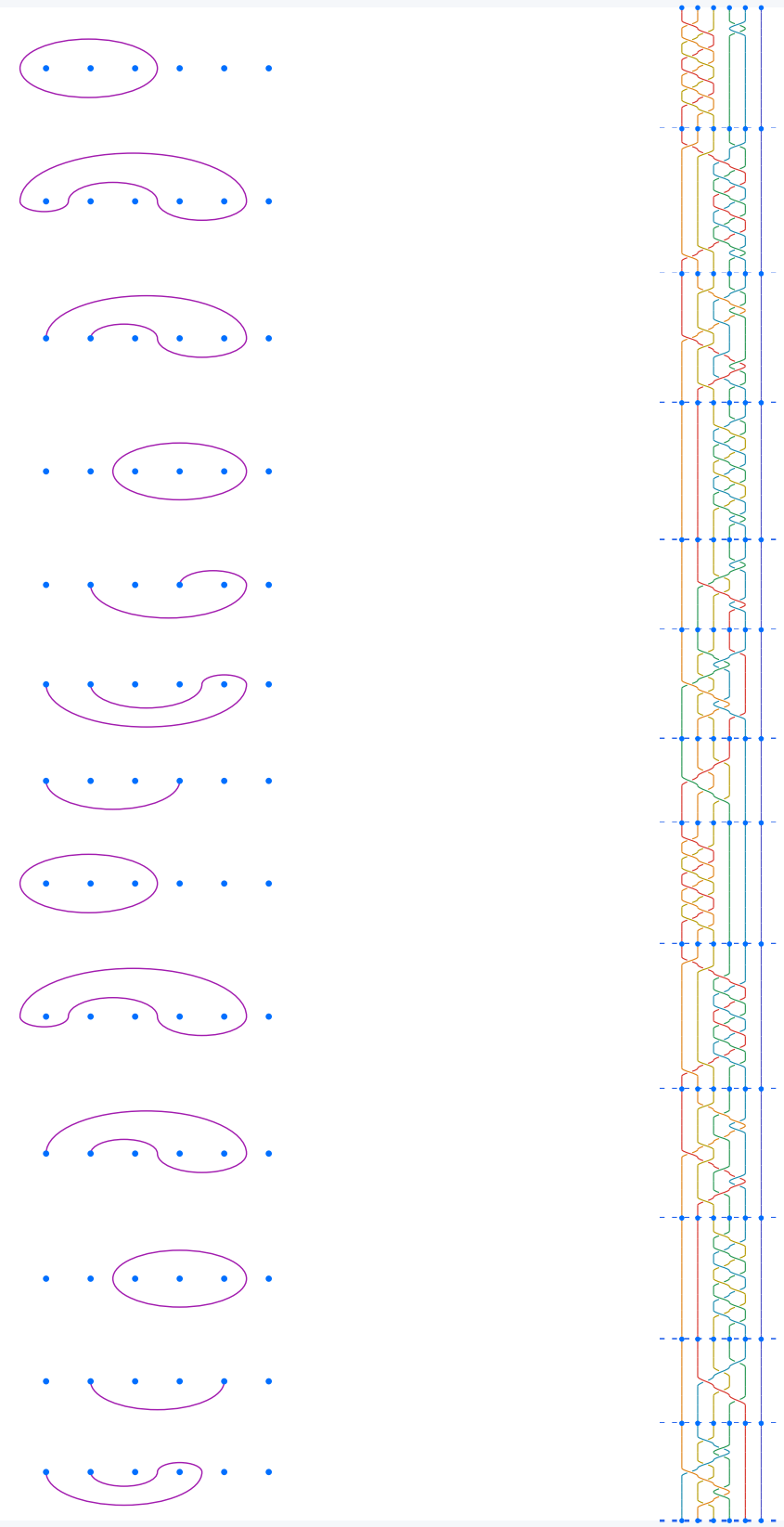
9. four_three_A.2
T3^2

10. four_three_A.3
H

11. four_three_A.4
T3^2

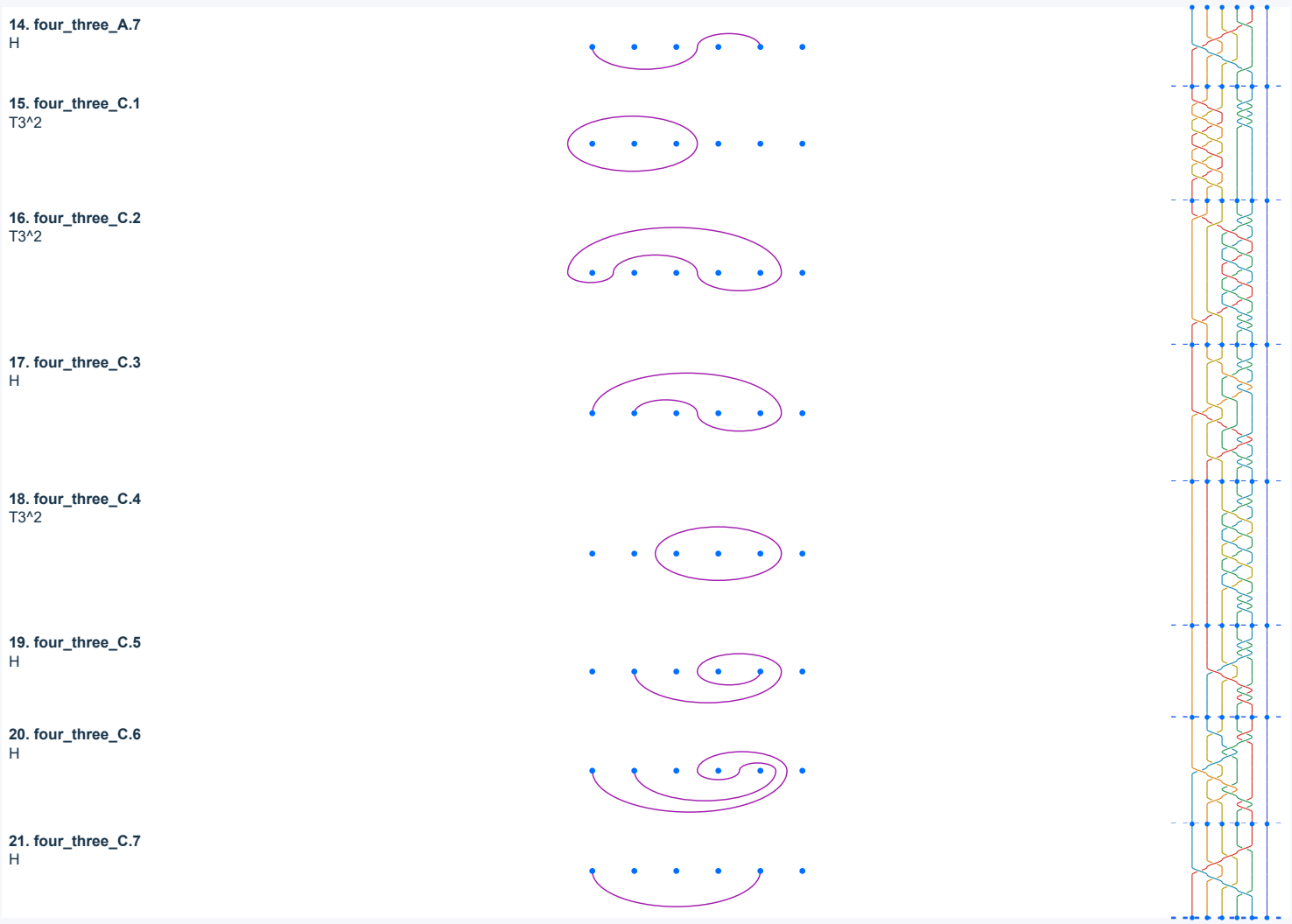
12. four_three_A.5
H

13. four_three_A.6
H



The explicit starting construction - continued

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.



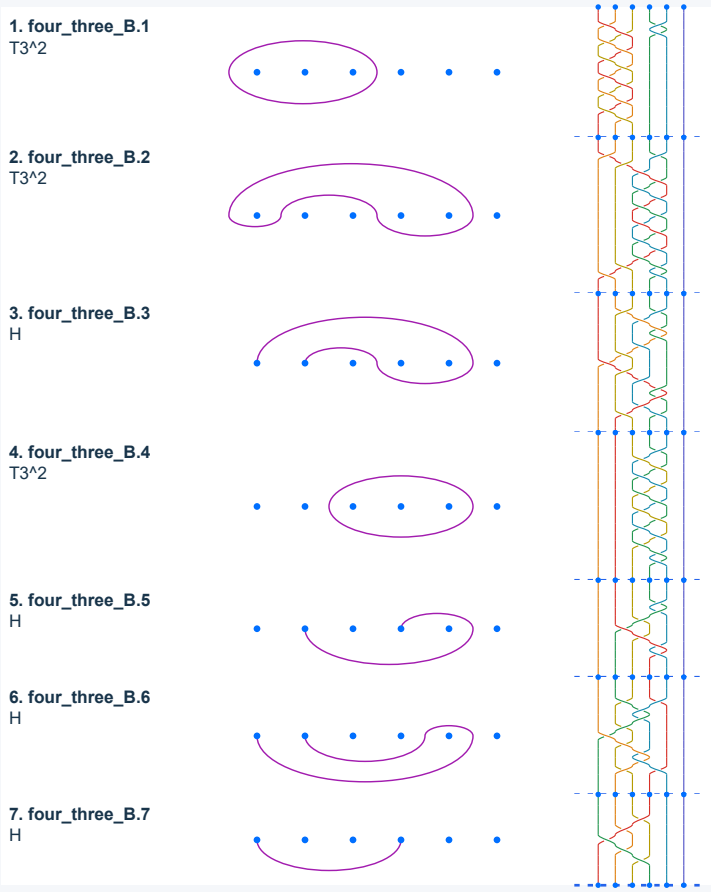
D - step 001a

Rotate block 1

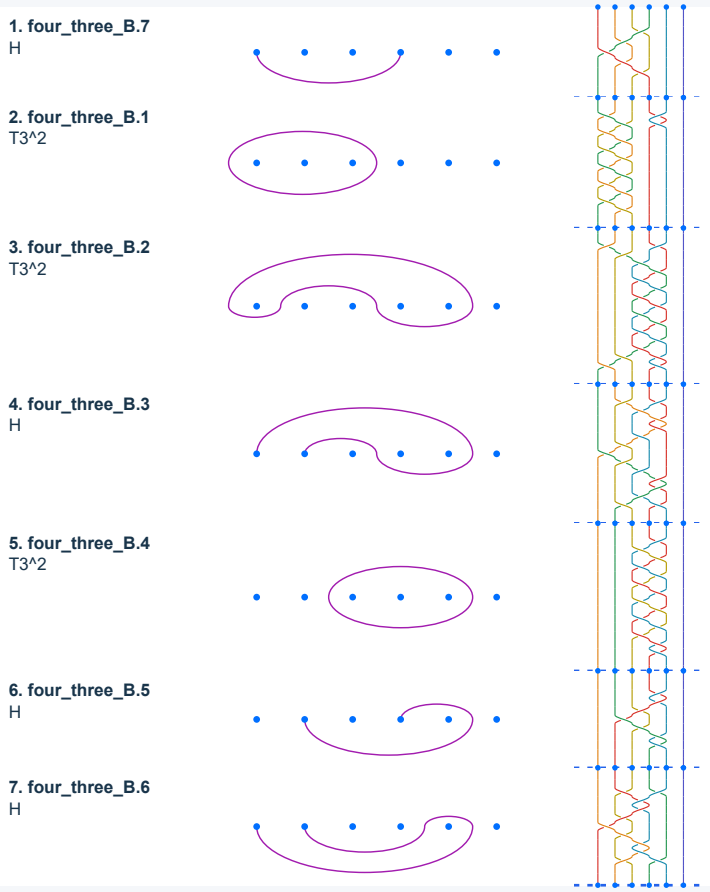
THREE INDEPENDENT DISK CYCLIC STARTS. Each acts on a different seven-factor block from the SAME input. All literal two-order certificates are retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown. Every parallel card starts from the SAME21-factor input; literal two-order equality is retained.

Before



After



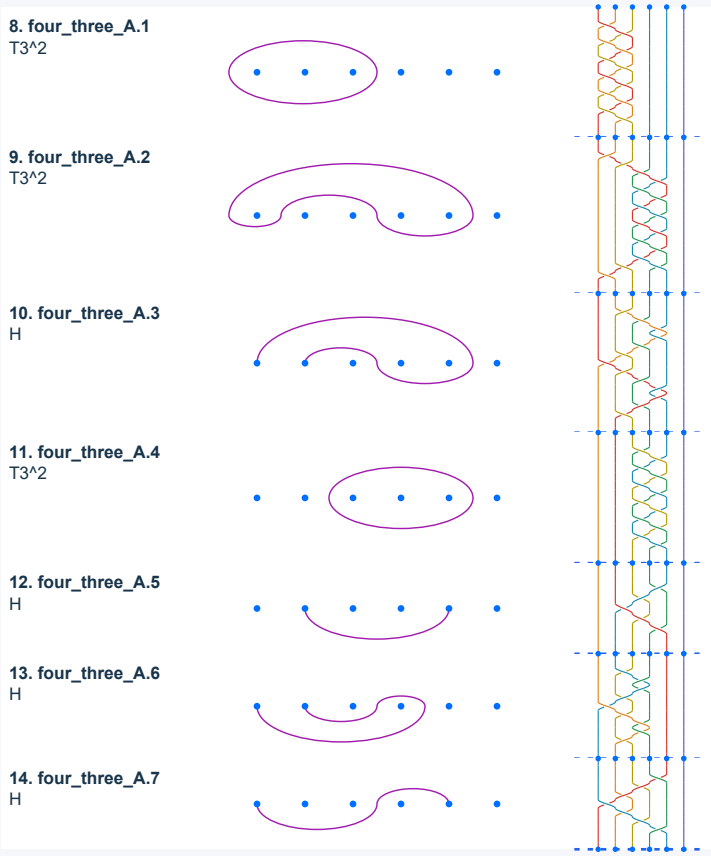
D - step 001b

Rotate block 2

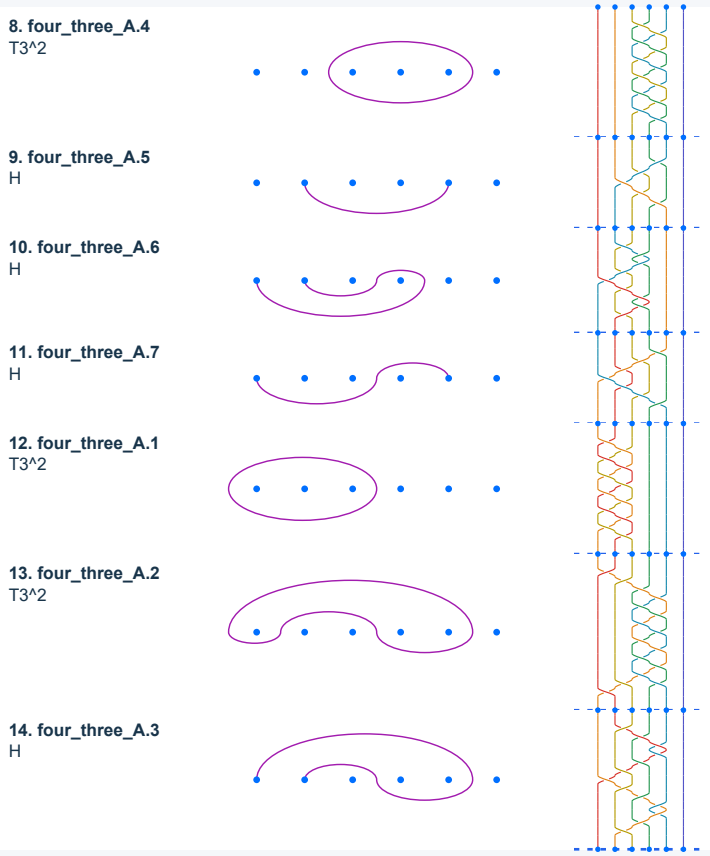
THREE INDEPENDENT DISK CYCLIC STARTS. Each acts on a different seven-factor block from the SAME input. All literal two-order certificates are retained.

7 unchanged displayed factors above; 7 below. Only the changed interval is shown. Every parallel card starts from the SAME21-factor input; literal two-order equality is retained.

Before



After



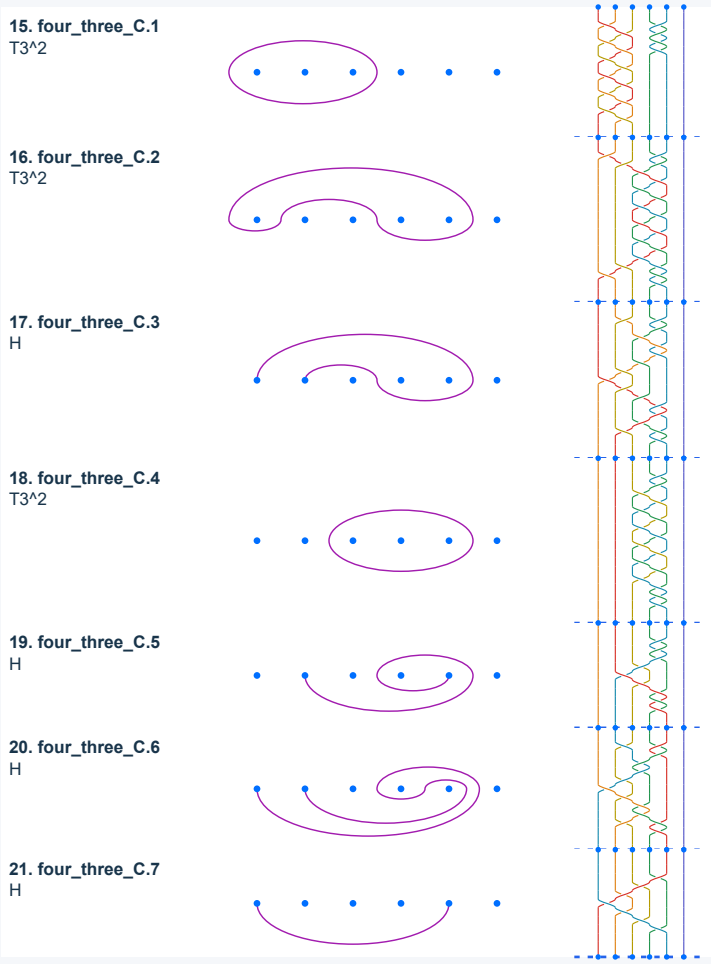
D - step 001c

Rotate block 3

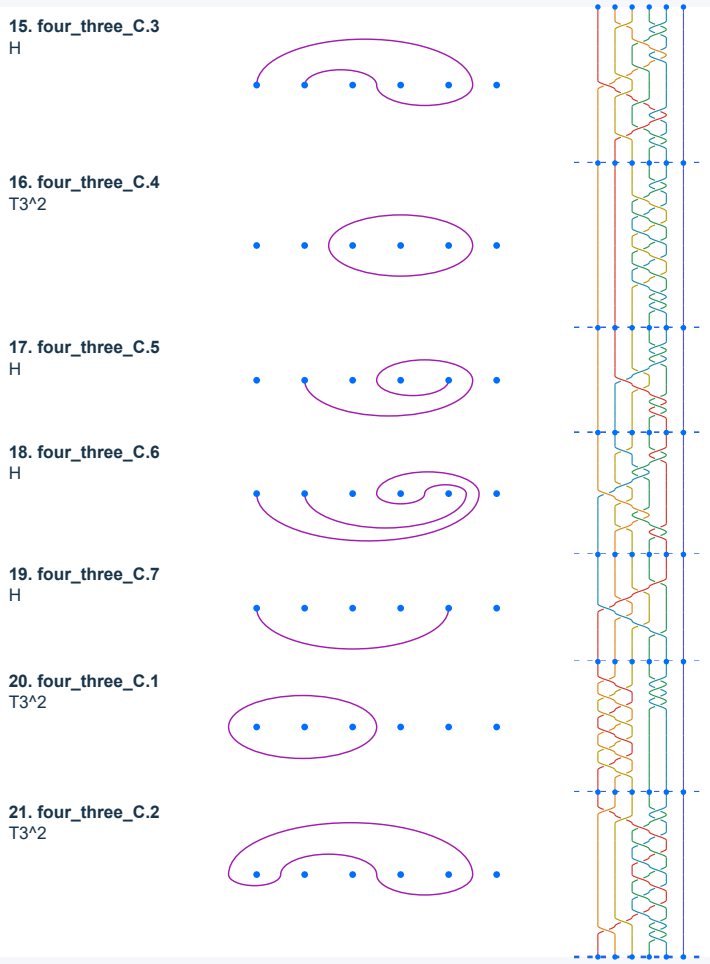
THREE INDEPENDENT DISK CYCLIC STARTS. Each acts on a different seven-factor block from the SAME input. All literal two-order certificates are retained.

14 unchanged displayed factors above; 0 below. Only the changed interval is shown. Every parallel card starts from the SAME21-factor input; literal two-order equality is retained.

Before



After



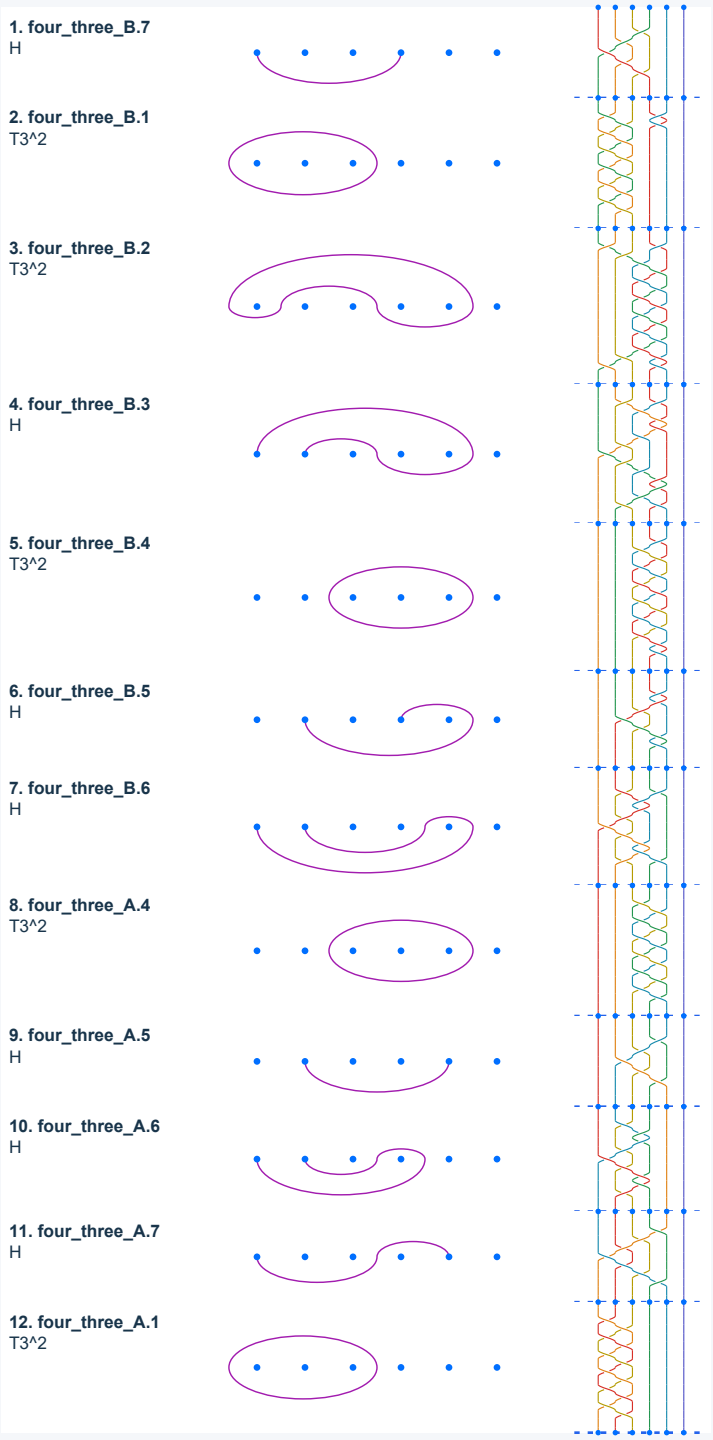
D - step 002

Rotate the FULL central21-factor product right by three

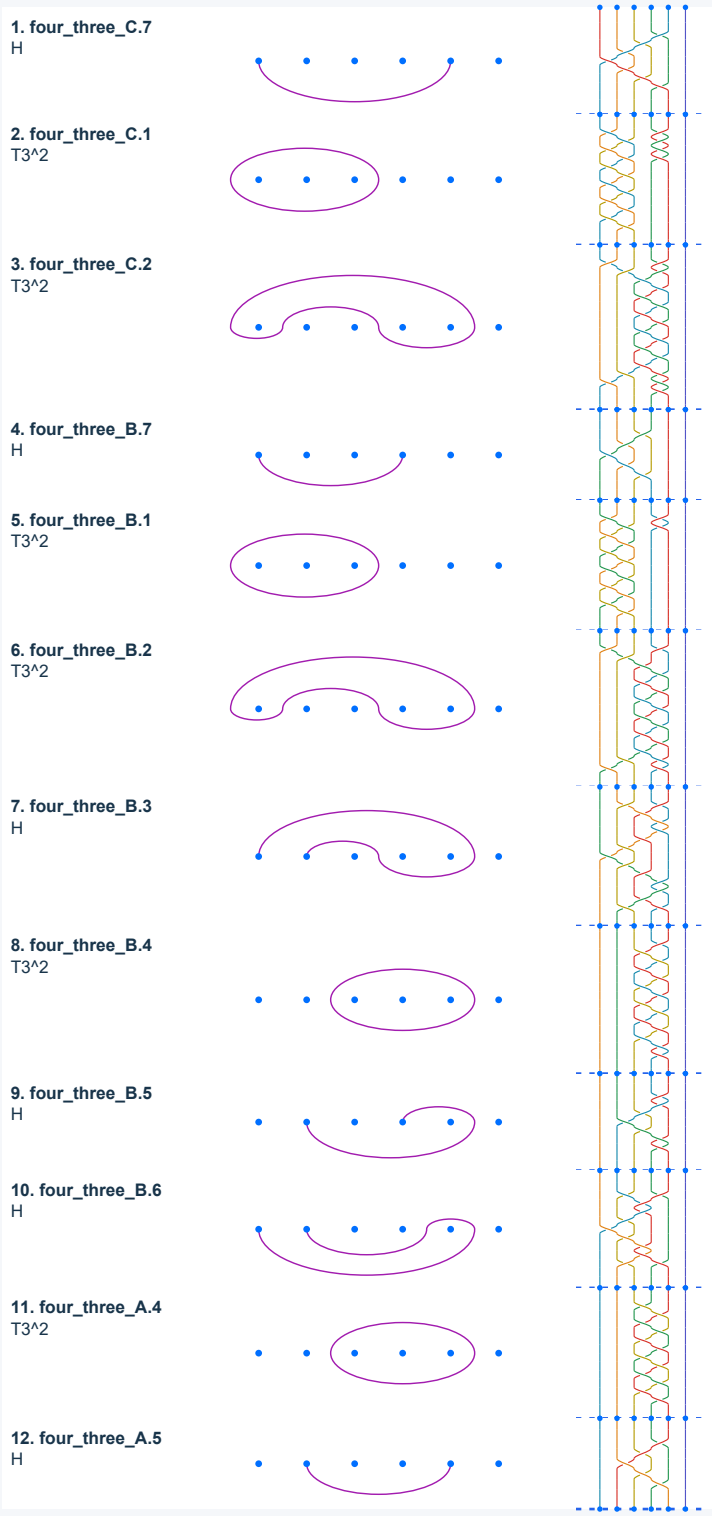
EXPLICIT CYCLIC START. Its full inverse-prefix common word is checked free first, with independently verified output invariance.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



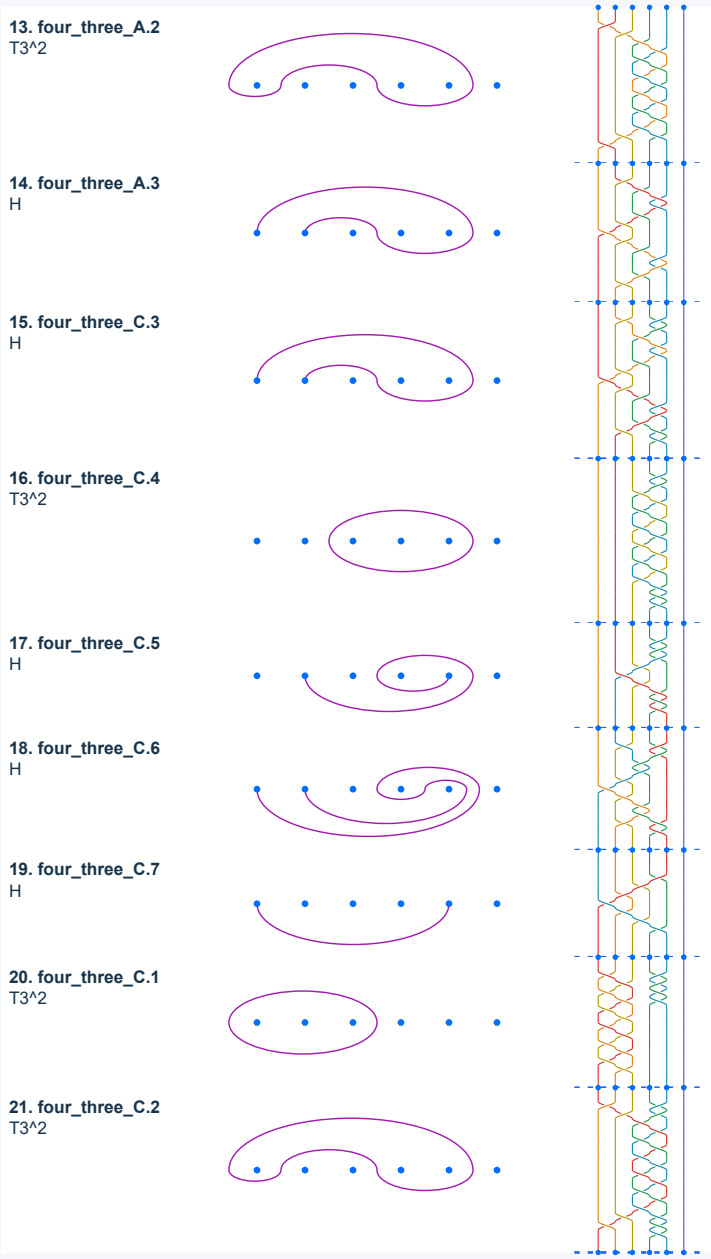
D - step 002 - continued

Rotate the FULL central21-factor product right by three

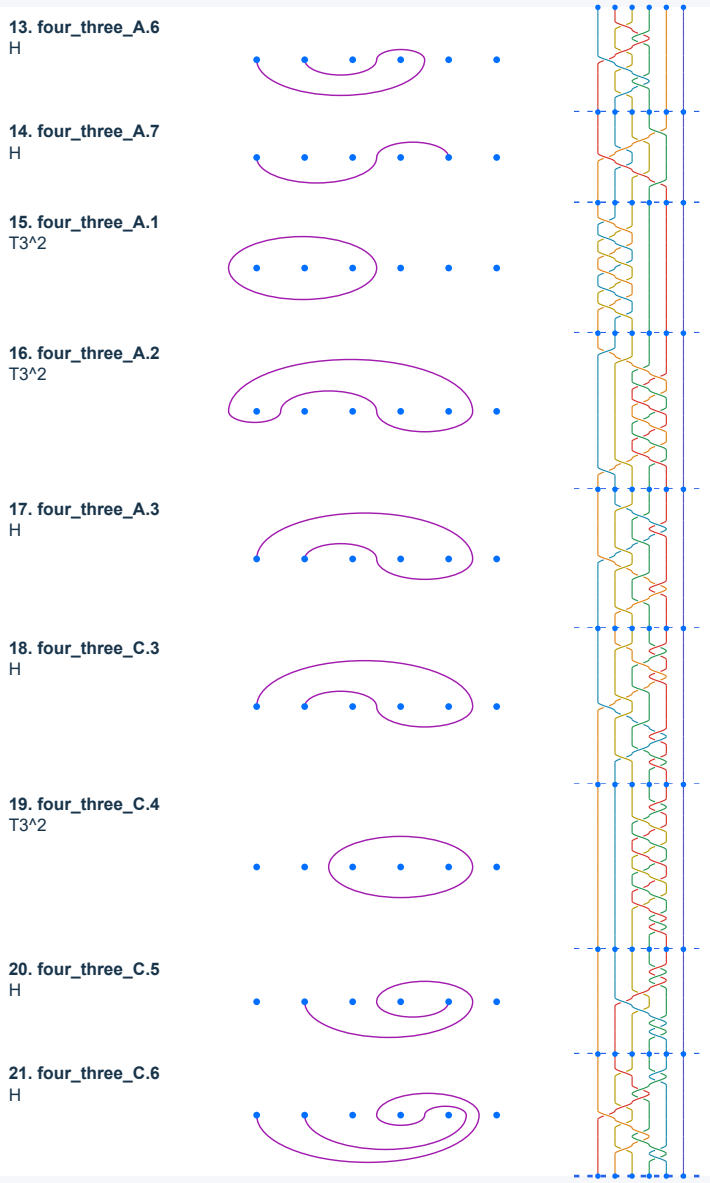
EXPLICIT CYCLIC START. Its full inverse-prefix common word is checked free first, with independently verified output invariance.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



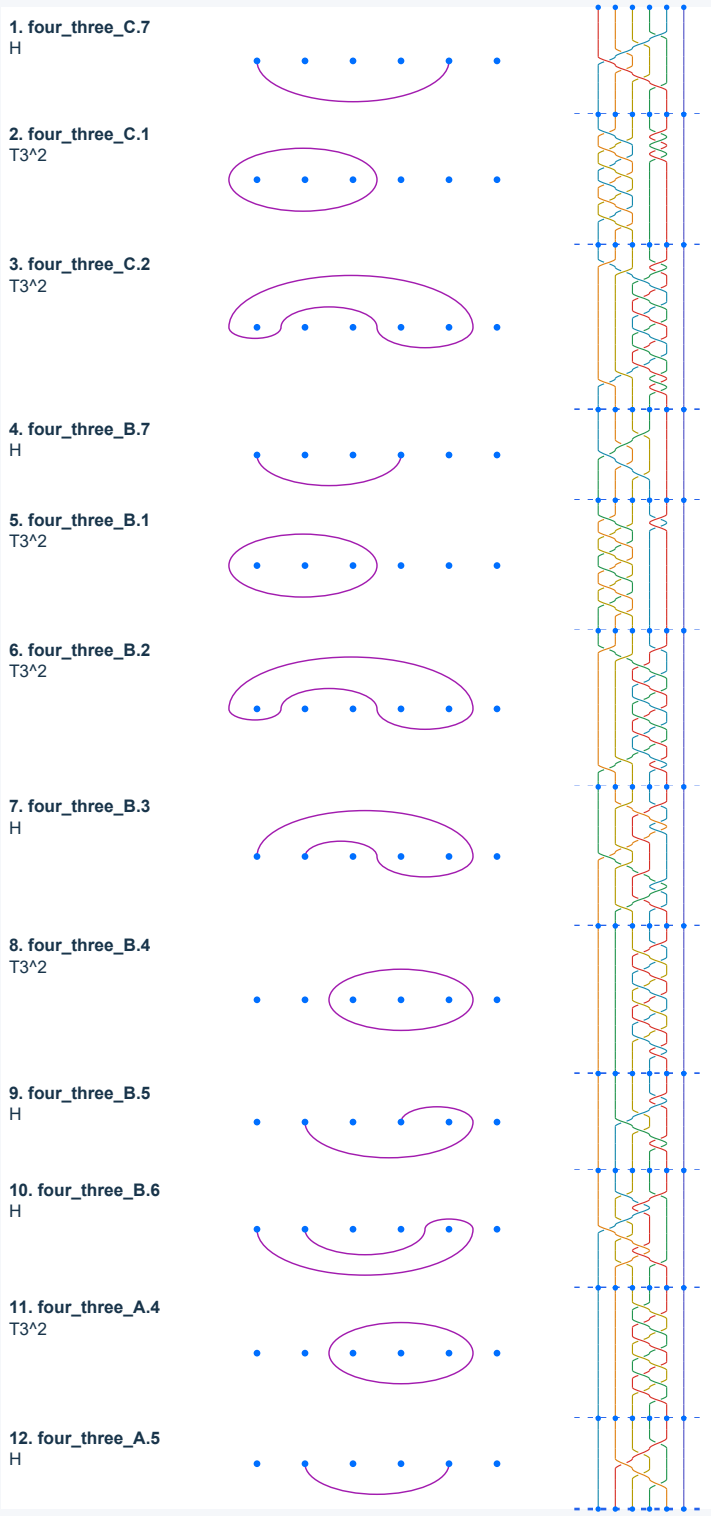
D - step 003

Perform the first two source pulls

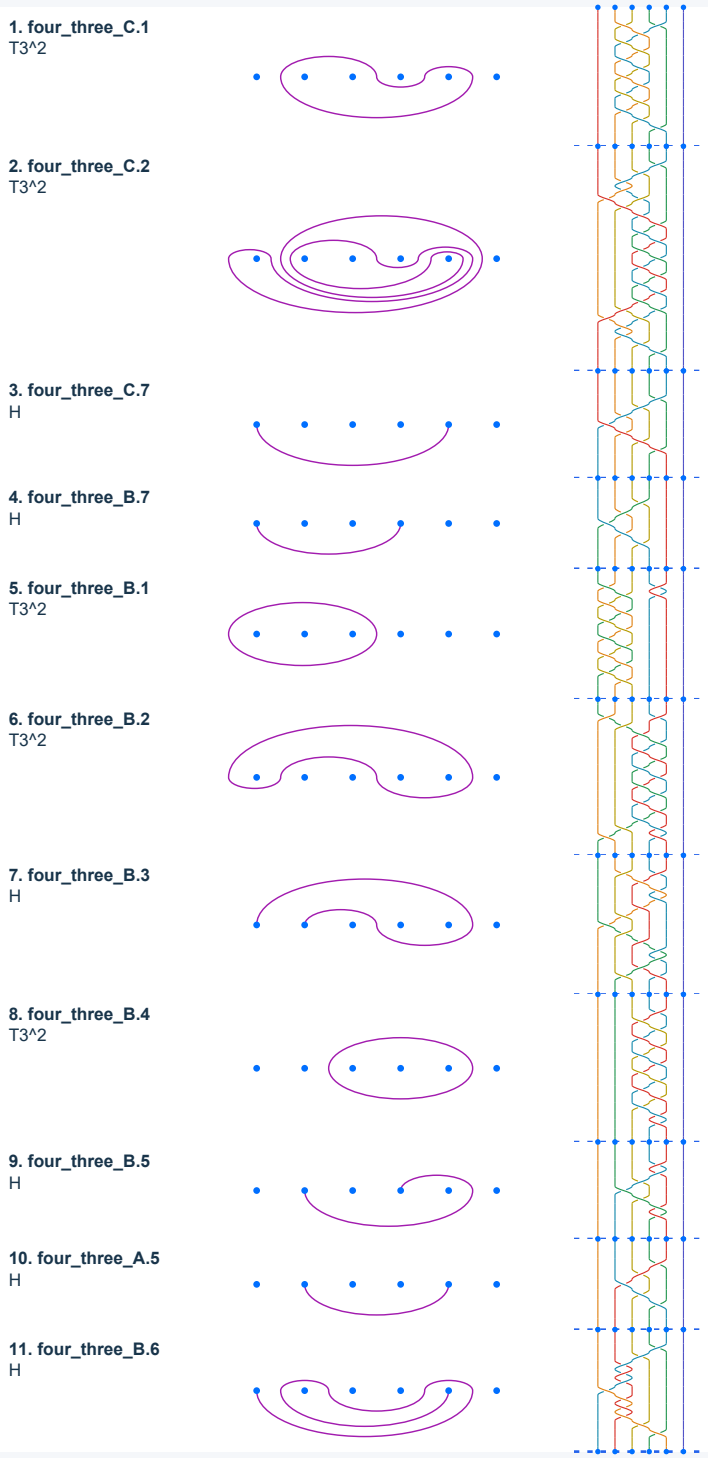
DISK TRANSPORT. Every adjacent crossing is raw-reconstructed and bounded20 before the complete carry is applied; all context changes remain recorded.

0 unchanged displayed factors above; 9 below. Only the changed interval is shown.

Before



After



D - step 003 - continued

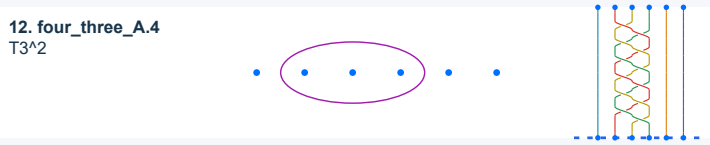
Perform the first two source pulls

DISK TRANSPORT. Every adjacent crossing is raw-reconstructed and bounded20 before the complete carry is applied; all context changes remain recorded.

0 unchanged displayed factors above; 9 below. Only the changed interval is shown.

Before

After



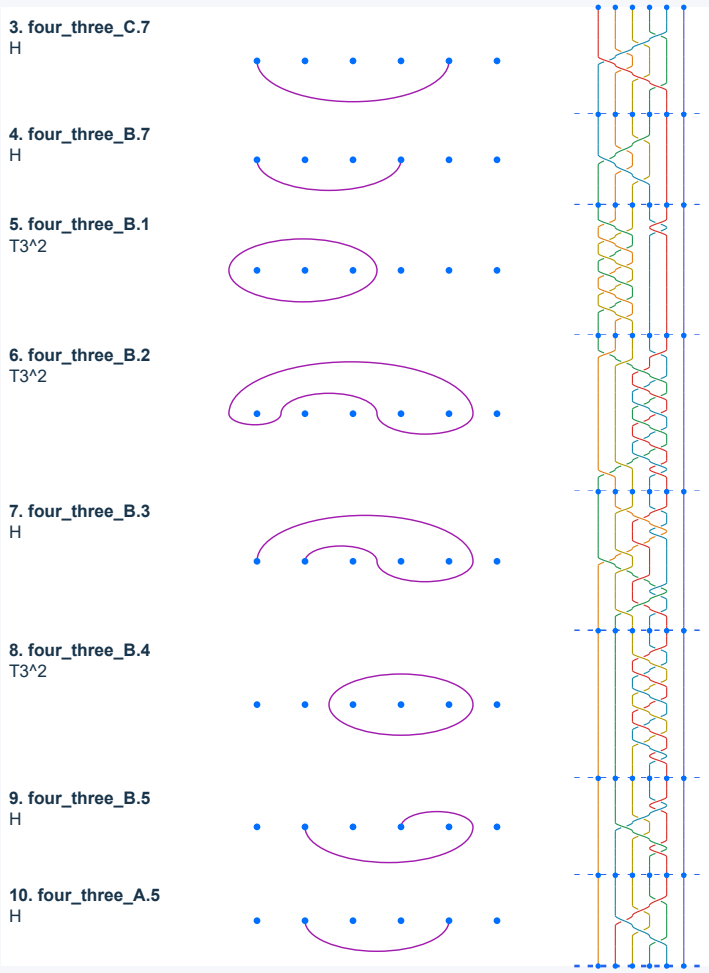
D - step 004

Perform the next two source pulls

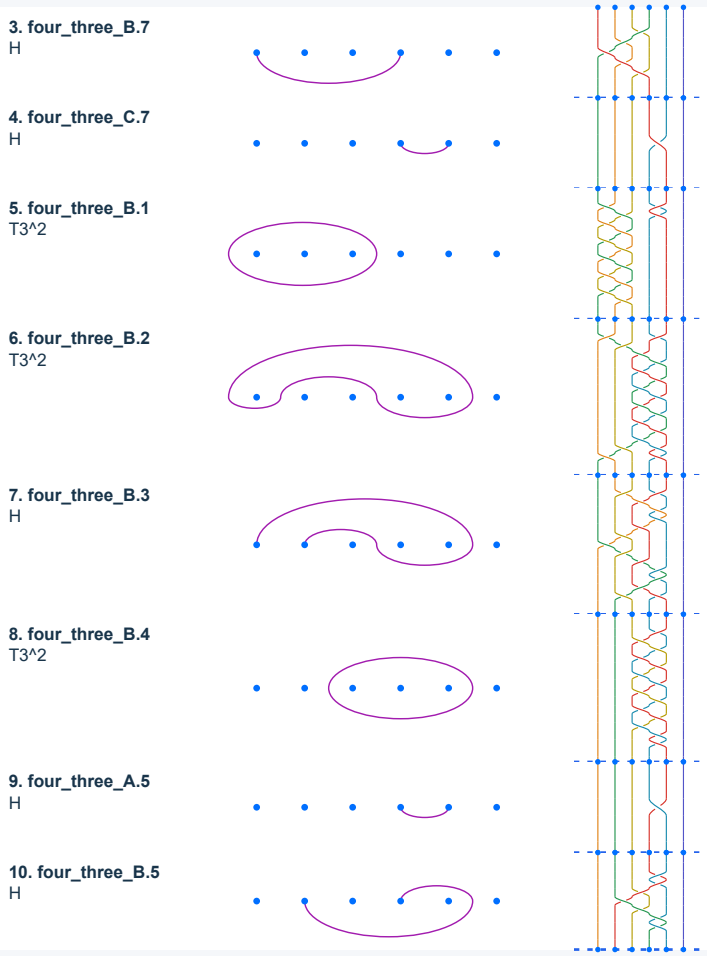
DISK TRANSPORT. Every adjacent crossing is raw-reconstructed and bounded20 before the complete carry is applied; all context changes remain recorded.

2 unchanged displayed factors above; 11 below. Only the changed interval is shown.

Before



After



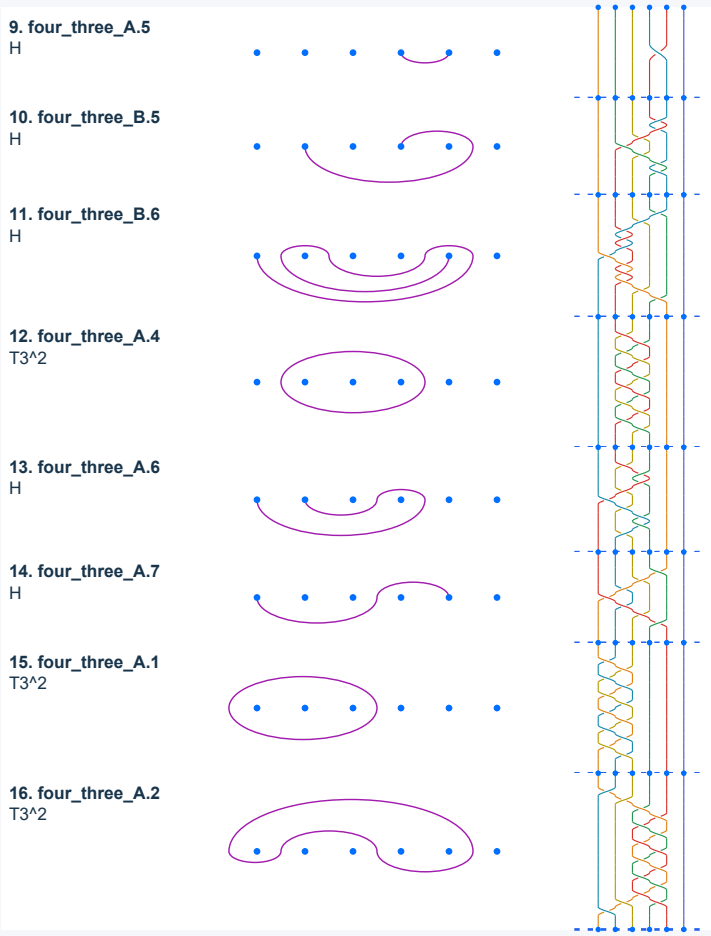
D - step 005

Carry the actual ninth half twist down to16

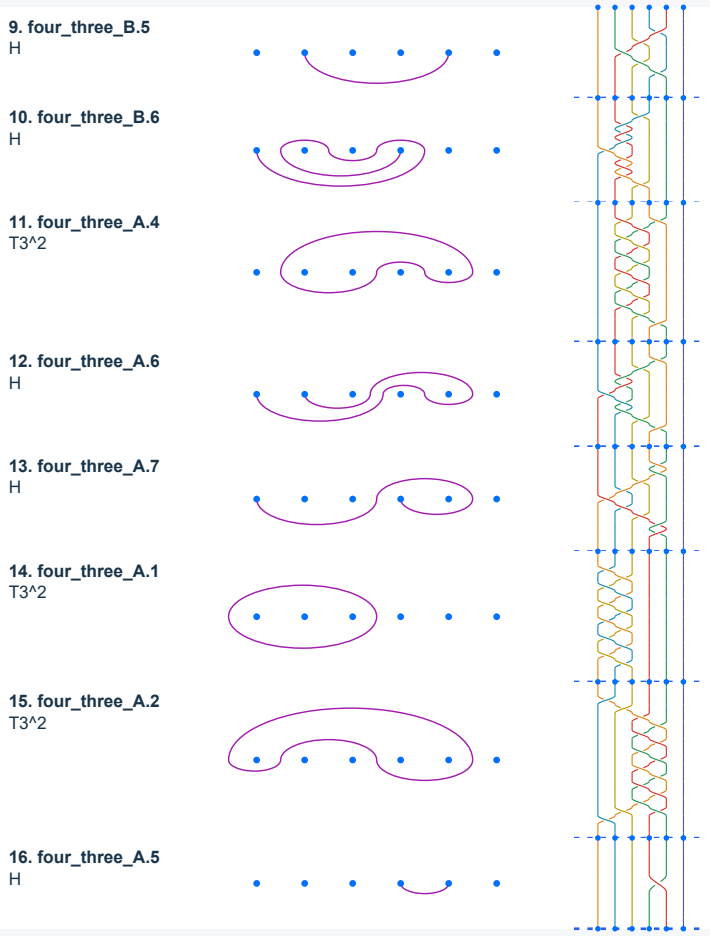
DISK TRANSPORT. Every adjacent crossing is raw-reconstructed and bounded20 before the complete carry is applied; all context changes remain recorded.

8 unchanged displayed factors above; 5 below. Only the changed interval is shown.

Before



After



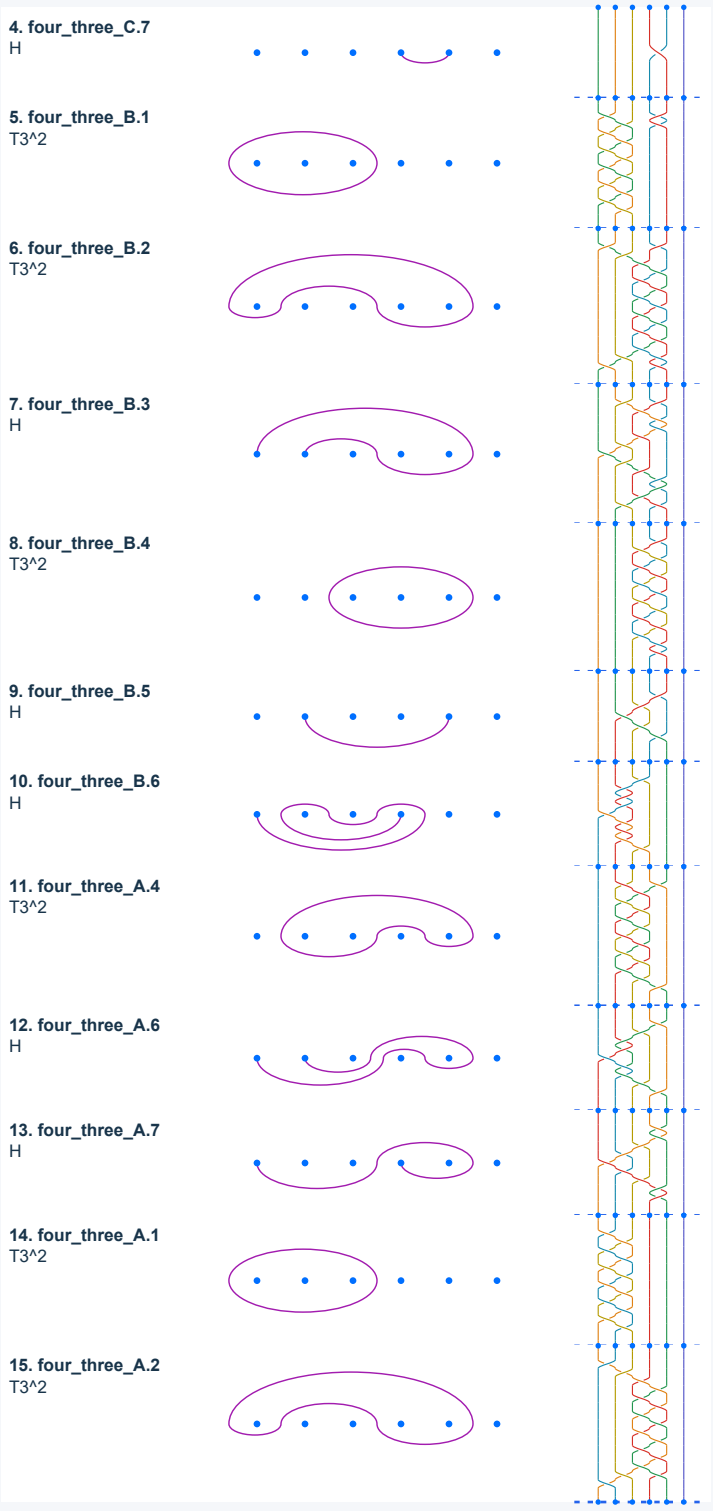
D - step 006

Carry the actual fourth half twist down to15

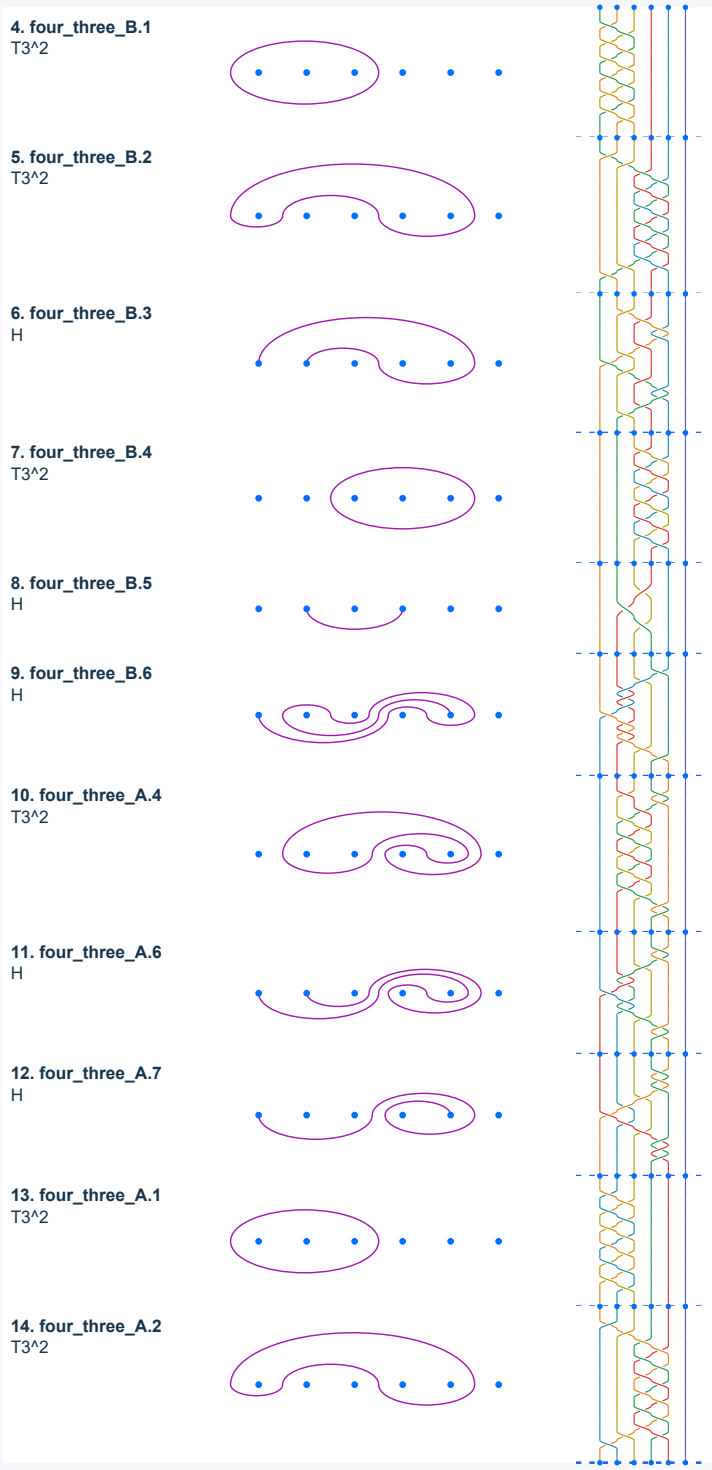
DISK TRANSPORT. Every adjacent crossing is raw-reconstructed and bounded20 before the complete carry is applied; all context changes remain recorded.

3 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



D - step 006 - continued

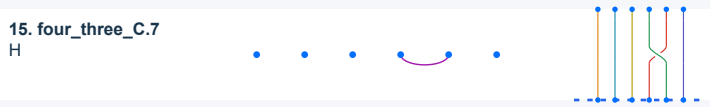
Carry the actual fourth half twist down to15

DISK TRANSPORT. Every adjacent crossing is raw-reconstructed and bounded20 before the complete carry is applied; all context changes remain recorded.

3 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before

After



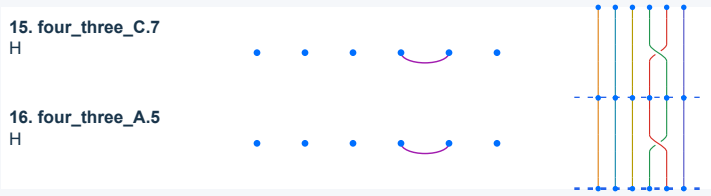
D - step 007

Combine the two matched merged.pair.one

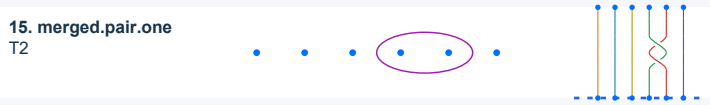
DISK COMBINE. The two supported twists are individually matched before their positive powers combine.

14 unchanged displayed factors above; 5 below. Only the changed interval is shown.

Before



After



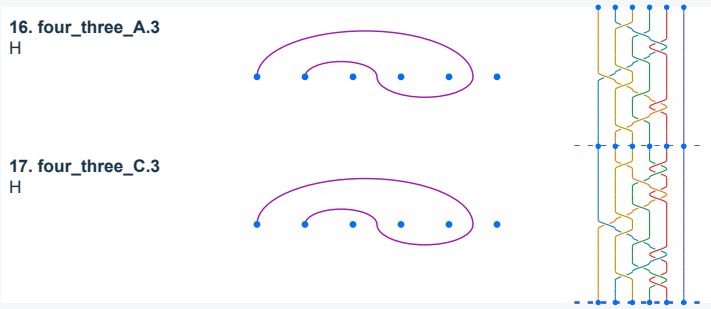
D - step 008

Combine the two matched merged.pair.two

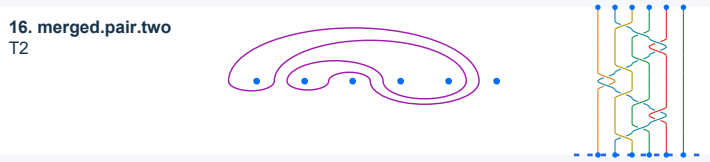
DISK COMBINE. The two supported twists are individually matched before their positive powers combine.

15 unchanged displayed factors above; 3 below. Only the changed interval is shown.

Before



After



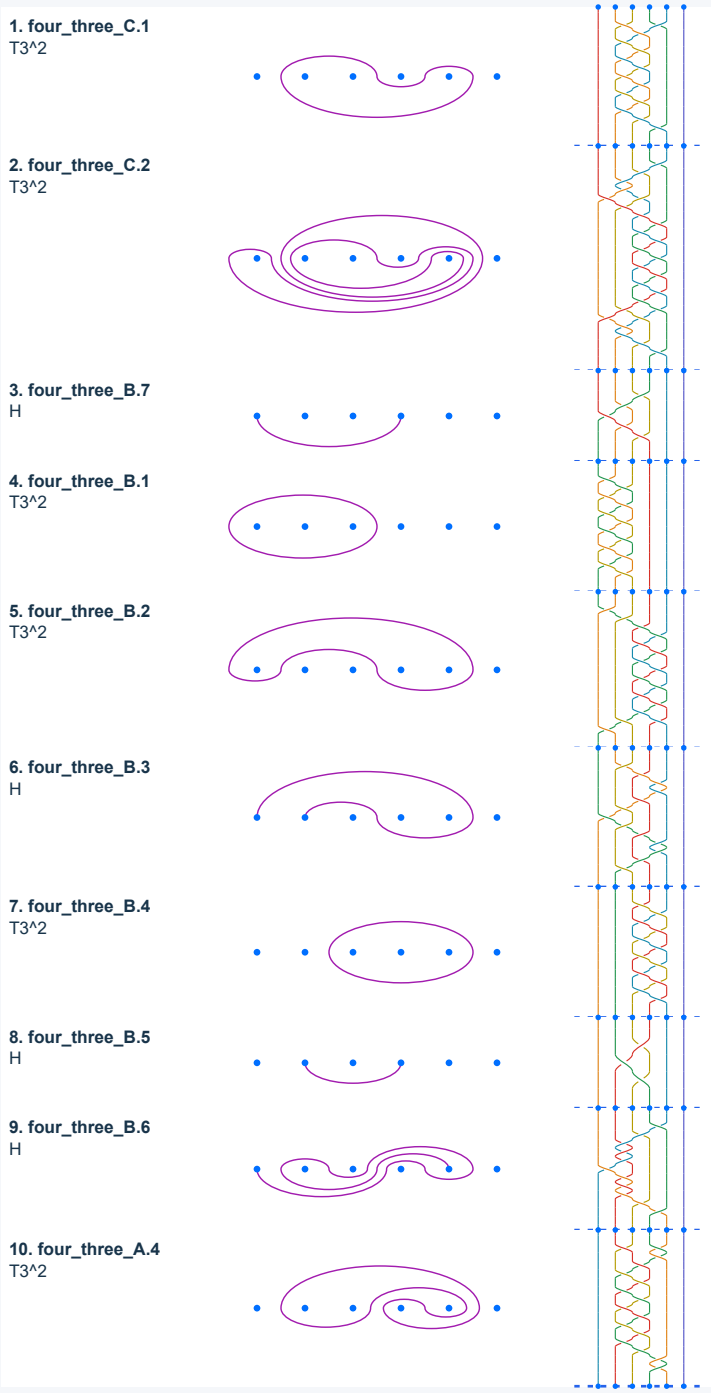
D - step 009

Embed the five-point construction; add the sixth point on the LEFT

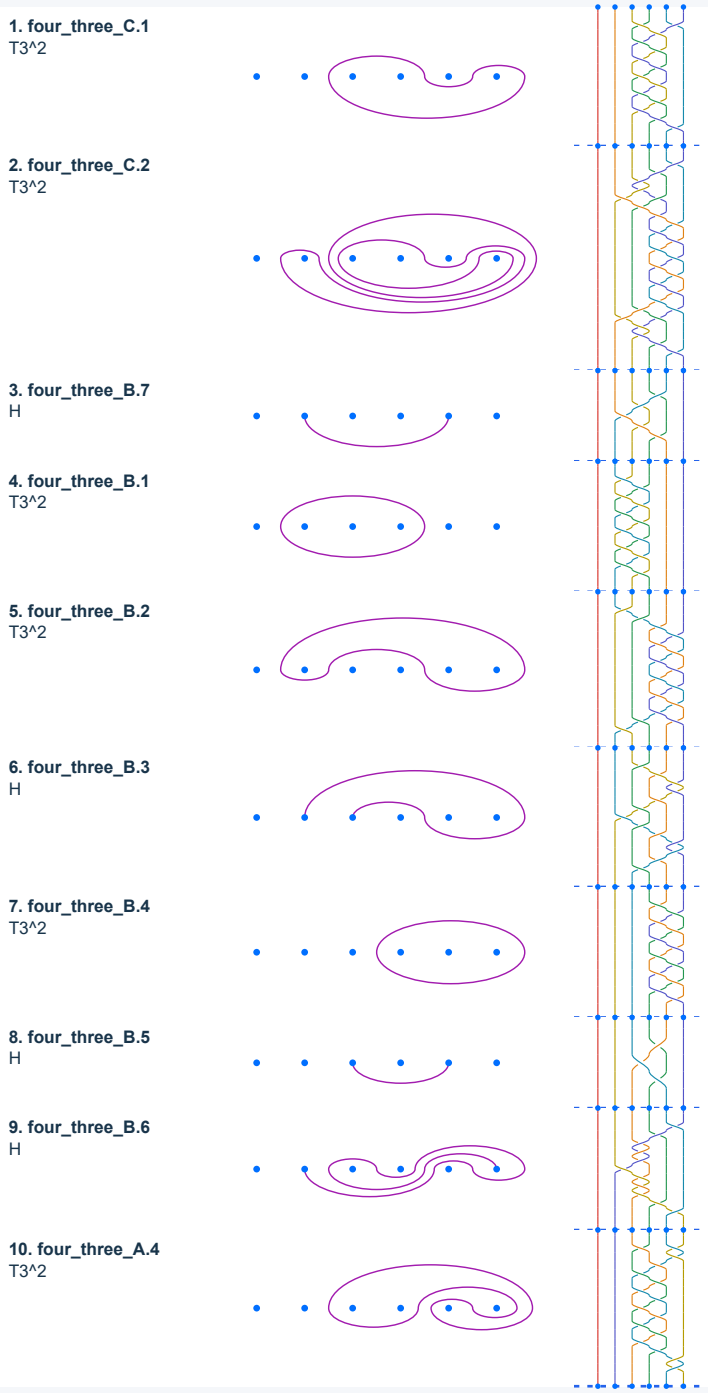
CONSTRUCTION / CONTEXT CHANGE. Add the sixth point on the LEFT and retain the complete embedded target. This is not an equality between B5 and B6, nor centrality of D5 in B6.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. The source and target contexts differ; this is labeled as a construction.

Before



After



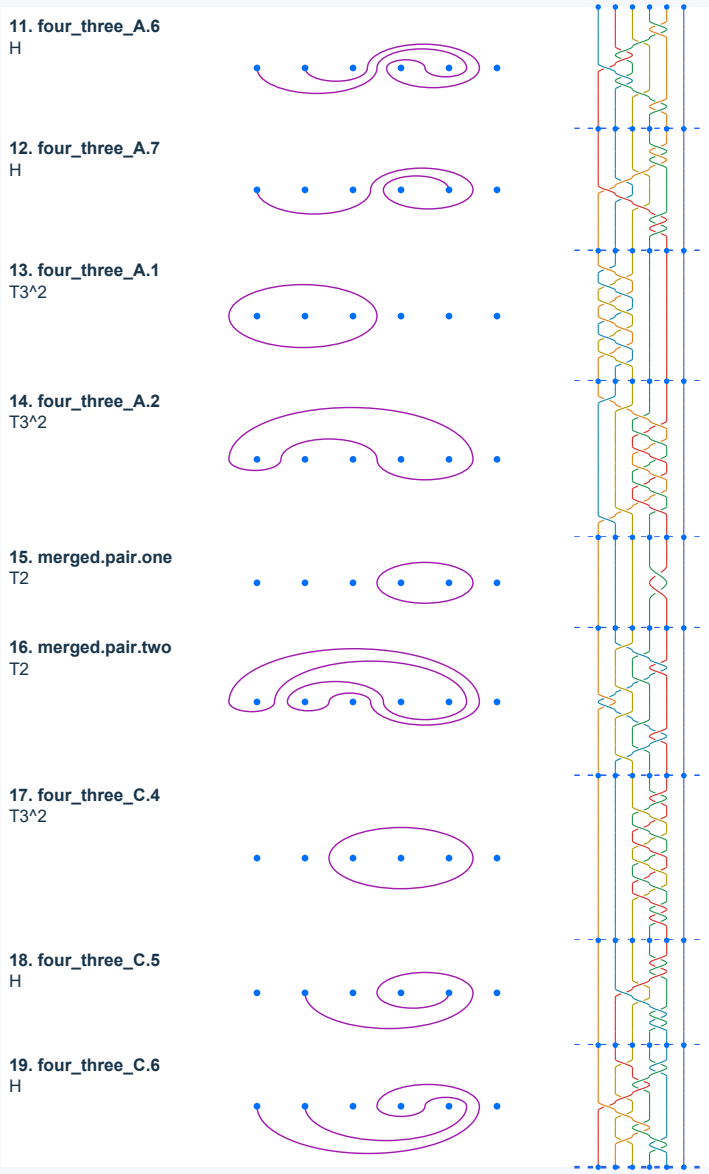
D - step 009 - continued

Embed the five-point construction; add the sixth point on the LEFT

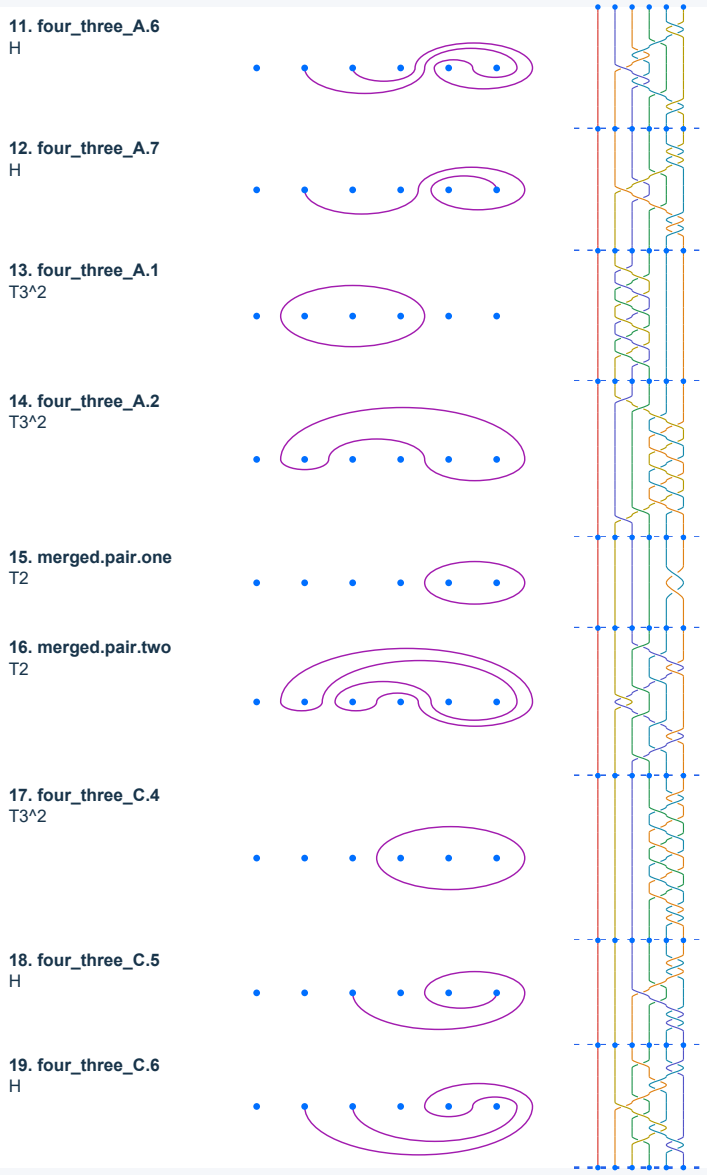
CONSTRUCTION / CONTEXT CHANGE. Add the sixth point on the LEFT and retain the complete embedded target. This is not an equality between B5 and B6, nor centrality of D5 in B6.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. The source and target contexts differ; this is labeled as a construction.

Before



After



D - step 010

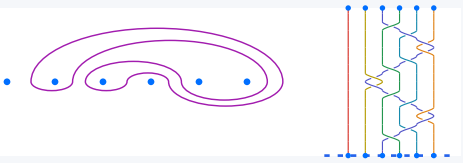
SPHERE REDRAW ONLY: the same actual pair curve from its complementary side

SPHERE REDRAW ONLY. The SAME actual curve, factor identity and power are shown from the complementary side. Full local AND complete-context disk discrepancies, marked-sphere action and chosen+I are retained. The following disk substitution has its own step.

15 unchanged displayed factors above; 3 below. Only the changed interval is shown.

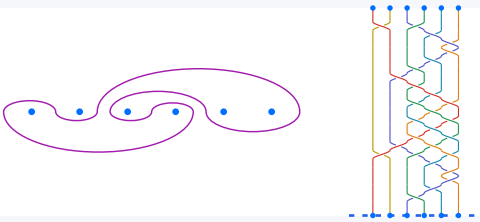
Before

16. merged.pair.two
T2



After

16. merged.pair.two
T4



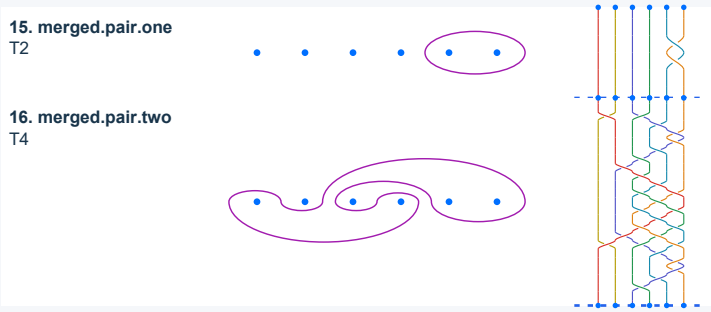
D - step 011

Apply the EXACT DISK half lantern after the separate redraw

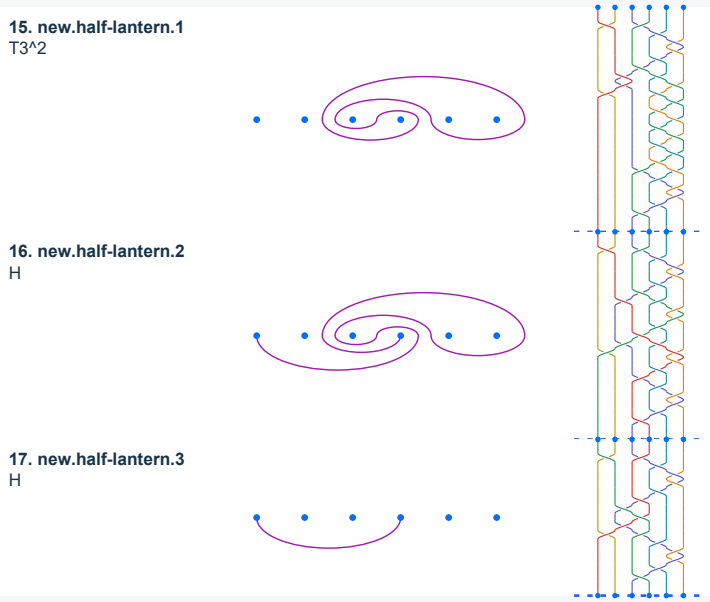
DISK SUBSTITUTION. Every ordered actual input/output and the full named local proof are checked. No sphere redraw or deletion is included.

14 unchanged displayed factors above; 3 below. Only the changed interval is shown.

Before



After



Complete final type (10,10) factorization

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

1. four_three_C.1
T3^2

2. four_three_C.2
T3^2

3. four_three_B.7
H

4. four_three_B.1
T3^2

5. four_three_B.2
T3^2

6. four_three_B.3
H

7. four_three_B.4
T3^2

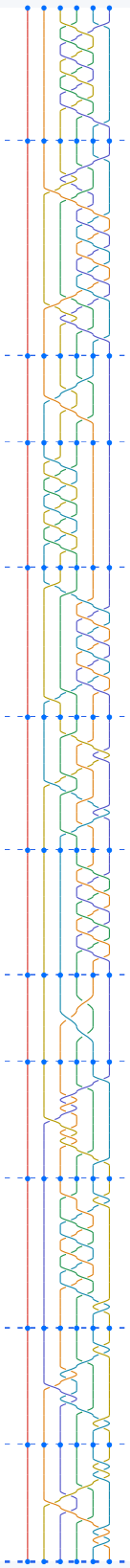
8. four_three_B.5
H

9. four_three_B.6
H

10. four_three_A.4
T3^2

11. four_three_A.6
H

12. four_three_A.7
H



Complete final type (10,10) factorization - continued

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

13. four_three_A.1
T3^2

14. four_three_A.2
T3^2

15. new.half-lantern.1
T3^2

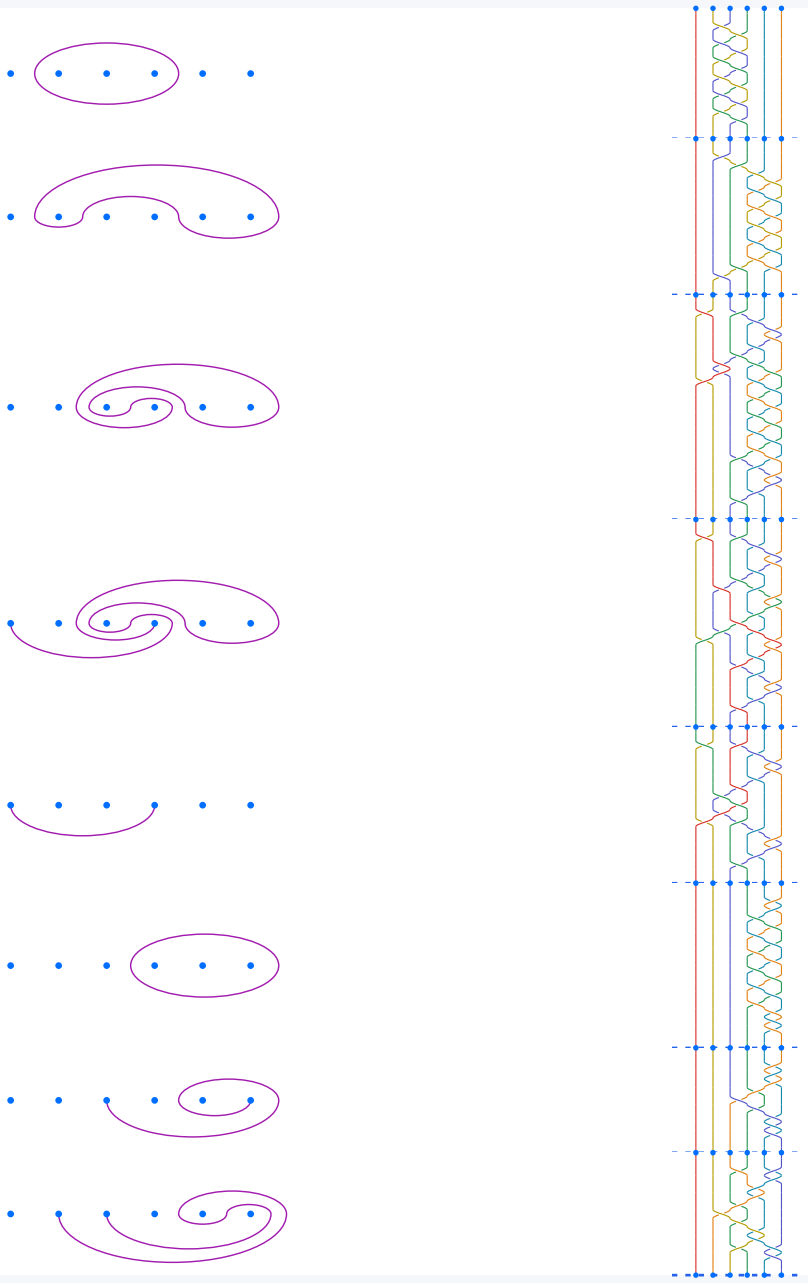
16. new.half-lantern.2
H

17. new.half-lantern.3
H

18. four_three_C.4
T3^2

19. four_three_C.5
H

20. four_three_C.6
H



Closure certificates and precise scope

The final product has the exact marked-sphere identity action and the separately checked chosen genus-two lift +I. This is not a claim of equality to the trivial disk braid, an unquotiented spherical-braid equality, a framed section lift, or an unproved identification with a particular literature curve chart.

Final chosen lift matrix: [[1, 0, 0, 0], [0, 1, 0, 0], [0, 0, 1, 0], [0, 0, 0, 1]]

Source-pinned full certificate: 10-10.json; graphical viewer: 10-10.html. Imported source manifest: ../fibration-sources/planar-mcg-2026-10-30/provenance.json. All sources, exact actual input/output matches, canonical PC daisies/half lanterns and parallel-copy corrections remain available with the repository.

FULL SPHERE redraw discrepancy

SPHERE REDRAW ONLY: the same actual pair curve from its complementary side

Only the SAME actual curve is redrawn. The whole surrounding factorization is included in the complete disk discrepancy below.

350 signed letters; full actual context retained

-5 3 4 3 4 3 4 3 4 3 4 -3 -4 2 3 4 5 4 5 4 5 4 5 4
5 4 5 -3 -2 4 3 3 4 5 -4 -3 2 3 4 2 3 2 3 2 3 2 3 2
3 2 3 -2 -3 4 5 4 5 4 5 4 5 4 5 4 5 3 2 -3 4 5 5 4
3 2 -3 -4 -5 -5 -4 3 4 5 4 5 4 5 4 5 4 5 4 5 -4 3 4 5
-4 -3 -3 -3 2 3 3 3 4 5 3 4 3 4 3 4 3 4 3 4 -3 2 3 3
4 5 -4 -3 2 3 4 -5 -5 -5 2 3 2 3 2 3 2 3 2 3 2 3 -2 -3
4 5 4 5 4 5 4 5 4 5 4 5 3 2 5 5 -3 4 5 5 4 3 2 2
1 2 -5 -4 -3 -5 -4 -3 -5 -4 -3 -5 -4 -3 -2 -1 -3 -4 -5 -5 -4 3 -5 -5
-2 -3 -5 -4 -5 -4 -5 -4 -5 -4 -5 -4 -5 -4 -5 -4 3 2 -3 -2 -3 -2 -3 -2 -3 -2
-3 -2 -3 -2 5 5 5 -4 -3 -2 3 4 -5 -4 -3 -3 -2 3 -4 -3 -4 -3 -4 -3
-4 -3 -4 -3 -5 -4 -3 -3 -3 -2 3 3 3 4 -5 -4 -3 4 -5 -4 -5 -4 -5 -4
-5 -4 -5 -4 -5 -4 -3 4 5 5 4 3 -2 -3 -4 -5 -5 -4 3 -2 -3 -5 -4 -5
-4 -5 -4 -5 -4 -5 -4 -5 -4 3 2 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 -4
-3 -2 3 4 -5 -4 -3 -3 -4 2 3 -5 -4 -5 -4 -5 -4 -5 -4 -5 -4 -5 -4 -3
-2 4 3 -4 -3 -4 -3 -4 -3 -4 -3 -4 -3 5